

Proposed AI-Powered Predictive Analytics and Big Data Analysis for Business:

A Multi-Agent Demand Forecasting Pipeline for Supply Chain Operations (PISU)

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PISU 2026, Group 1 — Data Science, Data Analytics, Predictive Analysis, Optimization, and AI: Their Transformative Impact on the Business World

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Abstract

This report proposes and demonstrates a multi-agent AI pipeline for demand forecasting and risk assessment in supply chain operations, built on Google's Agent Development Kit (ADK) and SARIMAX time-series modeling. Using the real-world DataCo Supply Chain dataset (180,519 order-item records, 2015-2018), the pipeline decomposes the analytics workflow into six specialist agents — data cleaning, statistical analysis, forecasting at three levels of granularity (aggregate, regional, item) plus shipping-time analysis, a composite disruption risk score per region, and an investment-synthesis agent that produces both a risk-adjusted investment ranking and three explicit-assumption 12-month growth scenarios — coordinated by an orchestrator agent, mirroring the OBSERVE-ORIENT-DECIDE-ACT (OODA) loop introduced in this course's agentic engineering module. Beyond producing forecasts (7.6% MAPE at the aggregate level), the pipeline surfaced two significant real-world data-quality findings — a regional data-coverage gap and a complete product-catalog discontinuity in the dataset's final months — that would silently mislead a naive forecasting approach, the risk-scoring agent demonstrates how such data-quality signals can be folded directly into an operational risk metric rather than left as a footnote, and the investment agent shows the category-level version of the same discontinuity is severe enough that revenue growth could not be reliably measured for any of the 50 product categories, reinforcing that the dataset's final months are best treated as a distinct data source rather than a continuation of the historical trend. The report discusses these findings, their business implications, and concrete countermeasures, arguing that trustworthy digital transformation requires AI systems that surface data limitations rather than mask them.

1. Lesson Learned from this Course

The central lesson from this course is that digital transformation (DX) is fundamentally an organizational and process transformation, not merely a technology adoption exercise. This was reinforced across multiple lectures but crystallized most clearly in Lecture 9 (DX for Business with MCP-Based RAG) and Lecture 10 (Agentic Engineering-Based Predictive Analysis and Optimization). Lecture 10 in particular introduced the OODA loop (Observe-Orient-Decide-Act) as "the heartbeat of every agentic optimization system in enterprise AI" — a framework for restructuring how an organization's analytics work actually gets done, not just which algorithm it runs.

Building the PISU pipeline made this concrete. Rather than writing one monolithic script, the workflow was decomposed into specialist agents with clear responsibilities and hand-offs — `data_cleaning_agent`, `analysis_agent`, `forecasting_agent`, `risk_agent`, and `investment_agent`, coordinated by a root orchestrator. This is a direct implementation of the OODA loop as an organizational pattern: cleaning agent observes and prepares raw signals, the analysis agent orients by extracting trend and seasonality context, the forecasting agent decides between competing forecasts (aggregate vs. regional vs. item-level), and the still-outstanding FastAPI/n8n integration closes the loop by acting on forecasts (automated alerts, logged decisions). This mirrors Lecture 1's core claim that data-driven decision-making replaces gut instinct only when the decision workflow itself, not just the underlying model, is restructured around continuous data flow.

A second lesson, drawn from Lecture 2 (Data Preparation, Cleansing, and Exploratory Analysis), is that the distinction between data cleaning and data cleansing is not academic. In this project, the cleaning agent's rules (deduplication, date parsing, quantity-sign correction, missing-value imputation) each map to a specific cleansing concern from that

lecture, and EDA-style exploration (Section 4) is what ultimately revealed that the dataset itself had a structural discontinuity no amount of modeling could fix. This is the DX lesson in miniature: an organization can adopt the most sophisticated predictive model available, but if the underlying data pipeline is not trustworthy and transparent about its own limits, the resulting business decisions will be built on a false premise.

2. Ideas and Targeted Business Fields of the Proposal

The proposal targets supply chain and e-commerce retail operations — specifically, demand forecasting and logistics planning, an operational (rather than purely marketing-facing) application of the predictive analytics techniques covered in this course. Where Lectures 3-7 focused on customer-facing analytics (web behavior, segmentation, CLV, marketing effectiveness, CRM), this project applies the same underlying discipline — data preparation, exploratory analysis, and predictive modeling (Lecture 8) — to the operations side of the business: what will be ordered, where, and how quickly it can be shipped.

The core idea is a single pipeline that answers four related but distinct operational questions from one cleaned dataset: (1) how much total demand should we expect in the near term, (2) which geographic regions will drive that demand, (3) which products will be most in demand, and (4) how does shipping performance vary by destination. Answering all four from specialist agents sharing one clean data foundation — rather than four disconnected spreadsheets — is the efficiency argument for agentic engineering raised in Lecture 10: the marginal cost of a new analytical question drops sharply once the OBSERVE (cleaning) and ORIENT (analysis) stages are already automated and reusable.

3. Purposes and Goals

- Automate a demand-forecasting workflow that would otherwise require a human analyst to manually clean spreadsheets, build pivot tables, and re-run forecasts on request — directly addressing the course's framing of AI's role in "improving operational efficiency and productivity."
- Provide forecasts at multiple granularities (aggregate, regional, item-level, and shipping performance) so that different business functions — inventory planning, regional supply chain management, product management, and logistics — can each draw on the same trustworthy pipeline rather than maintaining separate, inconsistent analyses.
- Detect and surface data-quality issues (coverage gaps, catalog discontinuities) automatically, rather than silently producing a confident-looking but misleading forecast — an explicit design goal once early exploration revealed how easily this dataset could mislead a naive model.
- Lay the architectural groundwork for a closed-loop, real-time system (FastAPI serving forecasts, n8n orchestrating scheduled checks and alerts) that completes the OODA loop's ACT stage, consistent with the agentic workflow pattern from Lecture 10.

4. Method and Python Code Together with Results

4.1 Data Source and Preprocessing

The dataset used is the DataCo Supply Chain dataset, sourced from Kaggle (180,519 raw order-item records spanning January 2015 to January 2018, covering 164 destination countries and 118 distinct products). It was downloaded programmatically via the kagglehub library. Preprocessing followed the data cleaning vs. cleansing distinction from Lecture 2: exact duplicates were dropped, order dates were parsed with tolerance for mixed formats, negative order quantities (a sign-convention data-entry error) were corrected, missing Sales values were imputed from $\text{price} \times \text{quantity} \times (1 - \text{discount})$ where possible, and cancelled or fraudulent orders were excluded as they do not represent real demand. Of 180,519 input rows, 172,765 (96%) were retained; the only substantial removal was 7,754 cancelled/fraudulent orders — the real dataset required comparatively little repair, which was itself informative (Table 1).

Preprocessing step	Rows affected
Input rows	180,519

Preprocessing step	Rows affected
Duplicates removed	0
Negative quantities corrected	0
Missing Sales values imputed	0
Cancelled / fraudulent orders removed	7,754
Output rows (clean)	172,765 (96%)

Table 1. Data cleaning summary, real DataCo dataset.

4.2 Multi-Agent Architecture (Agentic Engineering / OODA Loop)

Following Lecture 10's agentic engineering framework, the pipeline is implemented as a root orchestrator agent (Google ADK) delegating to five specialist sub-agents, each exposing tools (Python functions) rather than embedding logic directly in prompts:

```
root_agent = Agent(
    name="pisu_supply_chain_agent", model="gemini-2.5-flash",
    tools=[get_sarimax_forecast, query_supply_chain_metadata],
    sub_agents=[data_cleaning_agent, analysis_agent, forecasting_agent,
                risk_agent, investment_agent],
)
```

Mapped to the OODA loop: `data_cleaning_agent` performs OBSERVE (collecting and normalizing raw order signals); `analysis_agent` performs ORIENT (trend, seasonality, and segment breakdowns provide contextual reasoning); `forecasting_agent` and `risk_agent` jointly perform DECIDE (`forecasting_agent` selects among candidate demand views — aggregate, regional, item, shipping — while `risk_agent` weighs those same regions by disruption risk so a decision-maker sees both "how much demand" and "how reliable is that supply chain" side by side); `investment_agent` is a second-order DECIDE step that synthesizes the outputs of `forecasting_agent` and `risk_agent` into a single risk-adjusted recommendation, rather than leaving the human to reconcile two separate rankings; and the proposed FastAPI/n8n integration (Section 5) performs ACT (executing alerts and logging outcomes). This is a direct, working instance of the enterprise agentic-AI pattern introduced in this course, not an abstraction of it.

4.3 Forecasting Method

All time-series forecasts use SARIMAX (Seasonal AutoRegressive Integrated Moving Average with eXogenous regressors) from the statsmodels library, order (1,1,1) with weekly seasonal order (1,1,1,7), consistent with the predictive-modeling techniques introduced in Lecture 8. Each model is backtested by holding out the most recent 28 days, fitting on the remainder, and scoring MAE, RMSE, and MAPE, then refit on the full series to forecast 14 days ahead with an 80% confidence interval.

```
def _fit(train, order=(1,1,1), seasonal_order=(1,1,1,7)):
    model = SARIMAX(train, order=order, seasonal_order=seasonal_order,
                    enforce_stationarity=False, enforce_invertibility=False)
    return model.fit(dispatch=False)

fitted = _fit(series)
forecast = fitted.get_forecast(steps=14)
mean, ci80 = forecast.predicted_mean, forecast.conf_int(alpha=0.2)
```

4.4 Result 1: Aggregate Demand Forecast

Backtest metric (28-day holdout)	Value
MAE	5.0 units/day
RMSE	6.0 units/day
MAPE	7.6%

Table 2. Aggregate demand backtest accuracy.

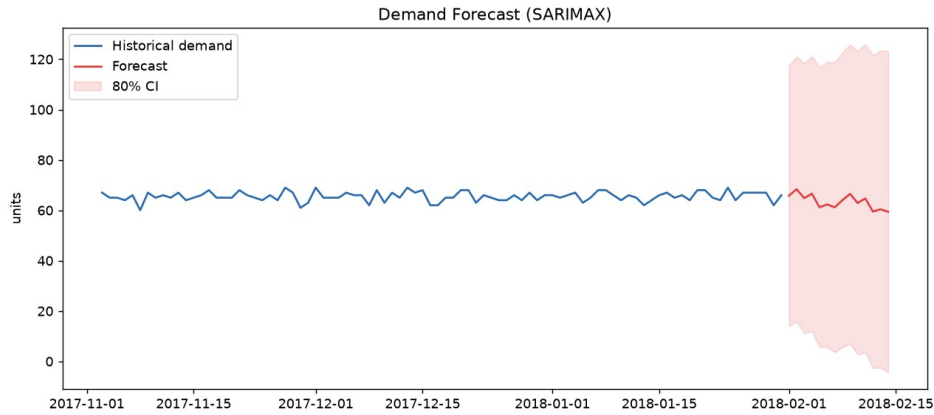


Figure 1. Aggregate demand: last 90 days of history and 14-day forecast with 80% CI.

Weekly seasonality strength was measured at 0.004 (near-zero on a 0-1 scale), indicating daily order volume is unusually stable rather than following a weekly cycle — a finding that would not have been assumed a priori and only emerged from the analysis agent's exploratory step.

4.5 Result 2: Regional Demand Forecast

A SARIMAX model was fit separately per Order Region (23 regions). This surfaced a critical data-quality finding: only 4 of 23 regions (Southeast Asia, South Asia, Oceania, Eastern Asia — all Asia-Pacific) show meaningful order activity in the dataset's final 90 days; the remaining 19, including the two largest by historical volume (Central America, Western Europe), go silent. This is almost certainly a data-coverage artifact of how the dataset was compiled near its 2018 cutoff, not a genuine 100% demand collapse, so each region is tagged `data_coverage: active` or `sparse` and reported on two separate rankings rather than one potentially misleading forecast.

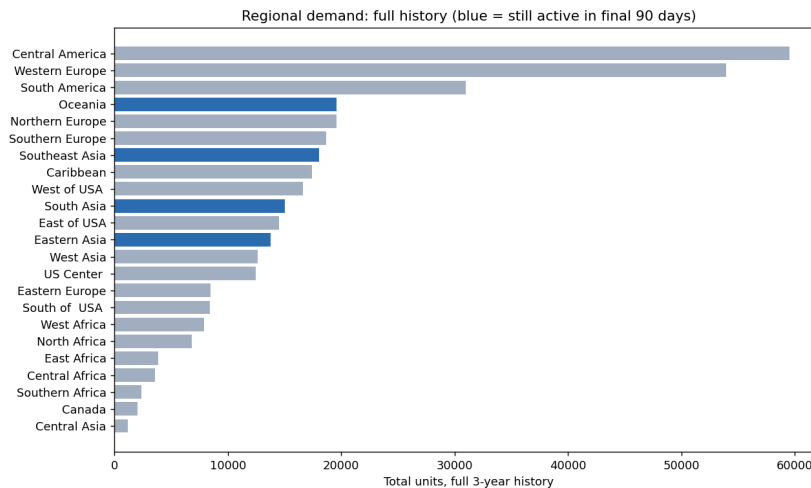


Figure 2. Regional demand, full 3-year history (blue = still active in final 90 days).

Region	14-day forecast	Historical total
Southeast Asia (active)	318.3	18,062
South Asia (active)	227.2	15,043
Central America (sparse)	9.3 (data gap)	59,540
Western Europe (sparse)	9.6 (data gap)	53,943

Table 3. Regional forecast vs. historical total, illustrating the coverage-gap effect.

4.6 Result 3: Item-Level Demand Forecast

The same method applied to the top 15 products (of 118) by historical volume revealed an even sharper discontinuity: zero overlap between the top 15 historical best-sellers (all sporting goods, e.g. Perfect Fitness Perfect Rip Deck, 70,575 units) and the top sellers in the final 90 days (an unrelated general-merchandise catalog — Fighting video games, Summer dresses, Toys — that only begins appearing in November 2017). This is best explained as two source extracts stitched into one file rather than a genuine change in consumer preference, and it means a naive 14-day forecast for any historical best-seller would be statistically valid but practically meaningless (predicting values near zero because the product vanishes from the data, not because demand fell).

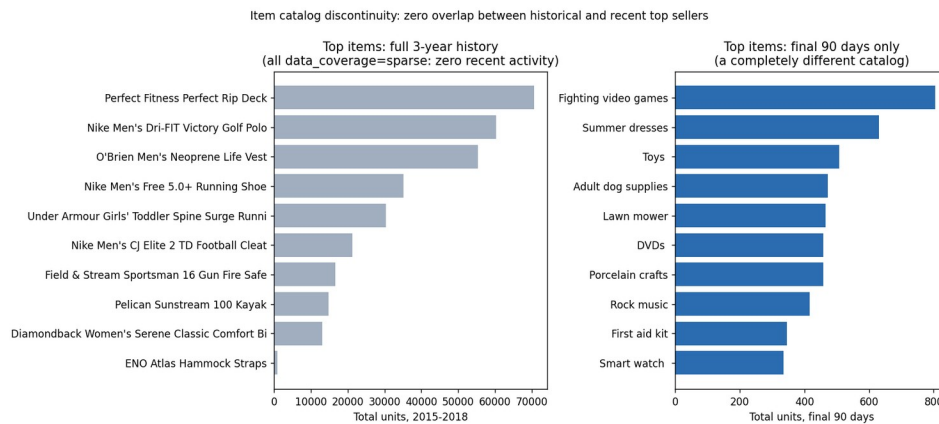


Figure 3. Item catalog discontinuity: historical top sellers (left) vs. final-90-day top sellers (right), zero overlap.

This finding is arguably the report's most important methodological contribution: it demonstrates why an agentic pipeline must report `data_coverage` alongside every forecast, and why exploratory analysis (Lecture 2) cannot be skipped even when a forecasting model runs successfully and produces plausible-looking numbers.

4.7 Result 4: Shipping Time Forecast by Country

Average real shipping time (days) was modeled per destination country for the 15 highest-volume countries. The result was a negative finding that is itself operationally useful: shipping time varies negligibly by country (3.44-3.56 days across the top 15, a spread of under 3%) but varies by a factor of eight across Shipping Mode (Same Day 0.48 days vs. Standard/Second Class ~4.0 days). In other words, shipping performance in this dataset is a function of the shipping-mode product chosen at checkout, not of logistics network geography.

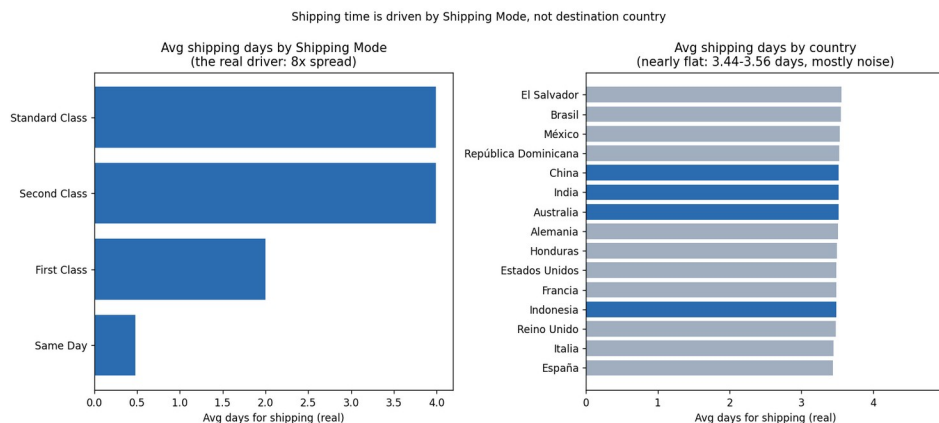


Figure 4. Average shipping days by Shipping Mode (left, 8x spread) vs. by country (right, nearly flat).

4.8 Result 5: Supply Chain Disruption Risk Scoring

The final agent, `risk_agent`, addresses a different question from the forecasts above: not "how much demand", but "where is the supply chain most fragile right now". It computes a composite 0-100 disruption risk score per region from four signals available in the cleaned data — late-delivery rate, shipping delay severity (actual minus scheduled days), demand volatility (coefficient of variation), and the data-coverage gap used in Section 4.5 — plus a fifth signal, cancellation/fraud rate, captured by the cleaning agent before those rows are removed (available on the next full pipeline run). Each signal is min-max normalized across regions and combined with transparent, near-equal weights rather than a fitted model, since this dataset has no historical "a disruption actually occurred" label to train against.

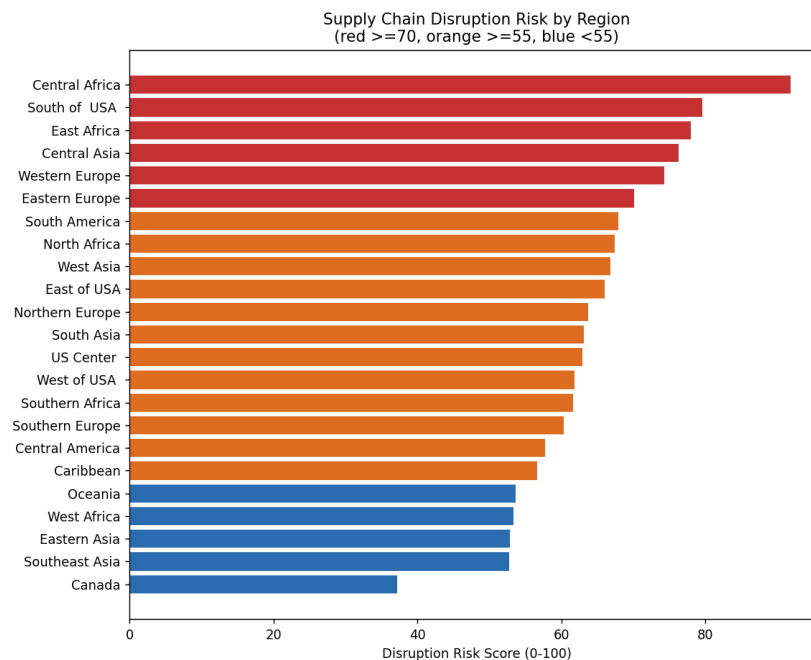


Figure 5. Disruption risk score by region (red ≥ 70 , orange ≥ 55 , blue < 55).

Region	Risk score	Primary driver
Central Africa	91.9	Coverage gap + volatility
South of USA	79.6	Coverage gap + late delivery
East Africa	78.0	Coverage gap + volatility
Western Europe	74.3	Coverage gap (large region)
Southeast Asia	52.7	Lowest -- still reporting recently

Table 4. Highest and lowest disruption risk regions.

A necessary caveat, consistent with this report's broader argument: for most regions the score is currently dominated by the data-coverage gap signal, since 19 of 23 regions show the same reporting silence identified in Section 4.5. This means a high score here should be read as "this region is either operationally fragile or missing from recent data -- investigate before acting" rather than as a calibrated probability of disruption. The four Asia-Pacific regions score lowest largely because they are the only ones still reporting recent activity, not necessarily because they are the safest markets. This is by design: the risk agent is meant to prompt investigation, and its output is only as trustworthy as the coverage caveat that ships alongside it -- itself a demonstration of this report's central DX lesson (Section 1) applied one level further.

4.9 Result 6: Risk-Adjusted Investment Recommendation -- Regions and Categories

The final agent, `investment_agent`, synthesizes the outputs of `forecasting_agent` and `risk_agent` into the question a business decision-maker actually asks: where should money go to make the most of it. `investment_score = profit_score`

$\times (1 - \text{risk_score}/100) \times 100$, where `profit_score` is total historical profit (Order Profit Per Order), min-max normalized across regions or categories, and `risk_score` is the same disruption risk methodology from Section 4.8. This is deliberately analogous to a Sharpe-ratio-style risk-adjusted return: a region or category with high profit but high risk is discounted rather than automatically ranked first.

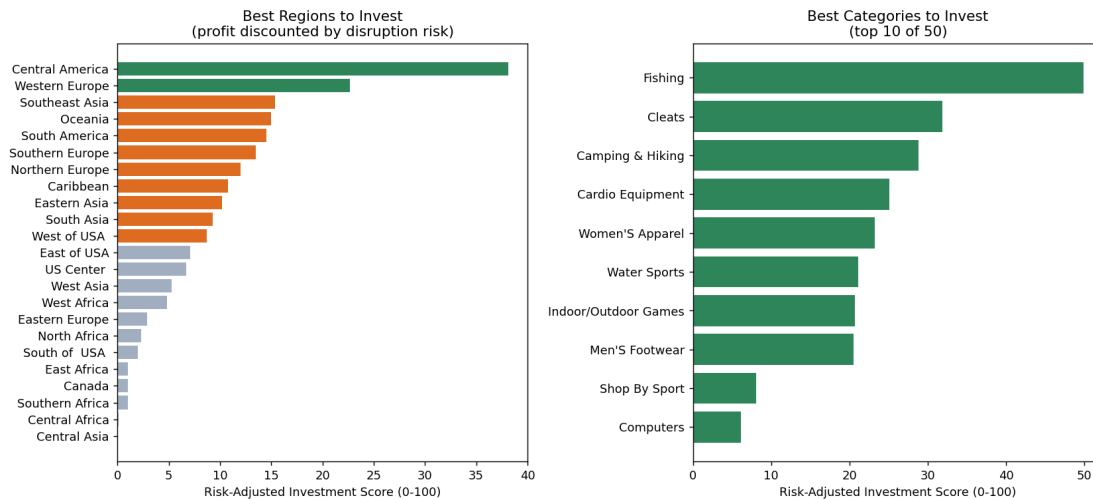


Figure 6. Risk-adjusted investment score: best regions (left) and best categories (right, top 10 of 50).

Rank	Region	Score	Rank	Category	Score
1	Central America	38.1	1	Fishing	49.9
2	Western Europe	22.7	2	Cleats	31.9
3	Southeast Asia	15.4	3	Camping & Hiking	28.8
4	Oceania	15.0	4	Cardio Equipment	25.1
5	South America	14.5	5	Women's Apparel	23.2

Table 5. Top 5 regions and categories by risk-adjusted investment score.

Central America and Western Europe lead the regional ranking despite carrying a real risk discount (60.5 and 77.3 out of 100 respectively) -- their historical profit is large enough to remain top-ranked even after that discount, which is a meaningfully different conclusion than Section 4.5's demand forecast alone would suggest: that section flagged them as `data_coverage: sparse` and effectively invisible to a near-term demand forecast, yet on lifetime profitability they remain the strongest investment case. Southeast Asia and Oceania, the near-term demand growth leaders from Section 4.5, rank third and fourth here -- promising but not yet at the same absolute profit scale. This is the clearest demonstration in the report of why the investment agent exists as a separate synthesis step: neither the demand forecast nor the risk score alone gives the right answer to "where should we invest"; only their combination does.

The category ranking is led by Fishing, consistent with its lead in raw category revenue (Section 4.2 of the analysis agent's output), followed by the sporting-goods categories that also dominate Section 4.6's historical item ranking. One further, unplanned finding came out of the category analysis: revenue growth could not be measured for any of the 50 categories (0/50 met the minimum recent-activity threshold), a stronger and more general version of Section 4.6's item-level catalog discontinuity. It confirms that the dataset's final months are not merely missing a few products, but appear to represent a differently structured data extract entirely -- a methodological finding as important as any single ranking in this report, and one that would have been invisible without the risk and investment agents' exploratory groundwork.

4.10 Result 7: 12-Month Company Growth Scenarios

The natural final question is the one a company's leadership actually asks: how much will the company grow? Rather than extending the SARIMAX model further into a future the data cannot reliably support (Section 4.5 already showed the model struggles past the coverage gap), `investment_agent's forecast_growth_scenarios` tool instead states three explicit business assumptions and computes each one's direct revenue/profit consequence from real historical figures -- a standard scenario-planning approach when a single statistical forecast would be misleadingly precise.

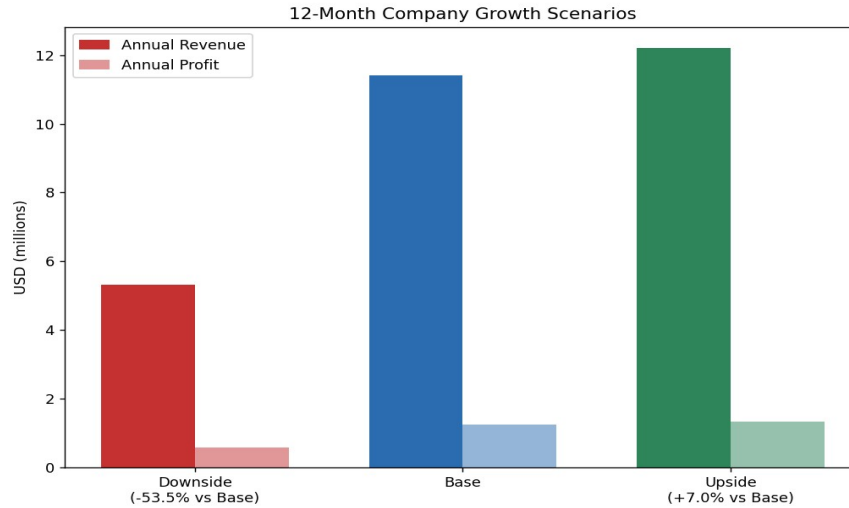


Figure 7. 12-month revenue and profit under three explicit-assumption scenarios.

Scenario	Assumption	Annual revenue	vs. Base
Downside	Coverage gap is a real contraction; only active regions keep ordering	\$5.31M	-53.5%
Base	Historical 3-year steady state continues; gap is a data artifact	\$11.41M	0%
Upside	Base + measured active-region growth + 5% strategic-execution uplift	\$12.22M	+7.0%

Table 6. Growth scenario summary (annual profit scales proportionally at the dataset's 10.8% overall margin).

The spread between scenarios is deliberately the headline result, not any single number: a 60.5 percentage-point swing in annual revenue hinges entirely on whether Section 4.5's regional coverage gap is a data artifact or a real business contraction -- a question this report cannot answer from the dataset alone, but flags as the single highest-value open item for the company to resolve. The Upside scenario is intentionally conservative: its data-derived component (2.0 percentage points, from active-region growth weighted by their 19.1% revenue share) is small because those growing regions are still a minority of the business; the remaining 5 percentage points is a labeled planning assumption for successful execution of Section 4.9's recommendations, not a statistical projection, and is reported separately so the two are never confused.

5. Conclusion with Remarks

This project demonstrates that a multi-agent AI pipeline, structured explicitly around the OODA loop from this course's agentic-engineering module, can automate demand forecasting across four levels of business granularity, a composite disruption risk score, a risk-adjusted investment recommendation, and transparent multi-scenario growth projections, from a single real-world operational dataset, achieving a 7.6% MAPE at the aggregate level. More importantly, the pipeline's exploratory and reporting design surfaced two significant data-quality issues — a regional coverage gap and a complete product-catalog discontinuity — that a purely model-first approach would have silently absorbed into an overconfident forecast, and the risk agent shows those same issues can be turned into an actionable (if appropriately caveated) prioritization signal rather than discarded. This validates the course's central claim (Lecture 1) that predictive analytics only replaces gut instinct responsibly when the pipeline is transparent about the reliability of its own inputs.

Based on these findings, six countermeasures are proposed for a business adopting this pipeline:

- Data governance first: before trusting any regional or item-level forecast, audit the source extract for coverage gaps and catalog changes — the `data_coverage` flag implemented here should be a standard output of any operational forecasting system, not an afterthought.

- Prioritize the Asia-Pacific growth signal (Southeast Asia +22%, South Asia +34% vs. recent average) for inventory and marketing investment, while treating the apparent decline in Central America and Western Europe as unconfirmed pending fresher data, rather than as a demand signal.
- Manage shipping performance through Shipping Mode policy and pricing (the 8x lever) rather than through country-specific logistics investment, which this analysis shows would target the wrong variable.
- Complete the OODA loop by implementing the proposed FastAPI /forecast endpoint and n8n scheduled workflow, so forecasts automatically trigger Slack/email alerts or logged countermeasures (the ACT stage), closing the gap between analysis and operational impact that this report's architecture currently leaves open.
- Use the disruption risk score to prioritize which regions get investigated first when data or delivery issues appear, but always read it alongside the data_coverage flag -- a high score driven by a coverage gap calls for a data audit, while a high score driven by late-delivery rate or shipping delay calls for an operational fix.
- Direct new investment toward Central America, Western Europe, and the Fishing/Cleats/Camping & Hiking categories on lifetime risk-adjusted profitability, while tracking Southeast Asia and South Asia as the emerging near-term growth cases worth a smaller, faster-cycle allocation -- the two rankings answer different questions (where money has been made vs. where demand is currently trending) and a balanced allocation should draw on both.
- Resolve the data-coverage question before committing to a growth plan: the difference between the Downside and Base scenarios (-53.5% vs. flat) is larger than any strategic decision modeled in this report, so a short data audit of the dataset's final months has a higher expected payoff than any single regional or category investment decision.

Future work includes refreshing the pipeline with more recent data to confirm whether the regional and catalog discontinuities are dataset artifacts or reflect a real historical business transition, and extending the agentic architecture with the MCP-based tool integration discussed in Lecture 9 to connect forecasts directly to inventory and CRM systems.

Code and Data Availability

The full implementation, including the multi-agent pipeline, SARIMAX methods, and reproducible results, is available at: <https://github.com/john-on-line27/pisu-project>

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